7R Mechanism Kinematics & Singularity Analysis Report

Computational Analysis

March 18, 2026

1 Kinematic Visualization

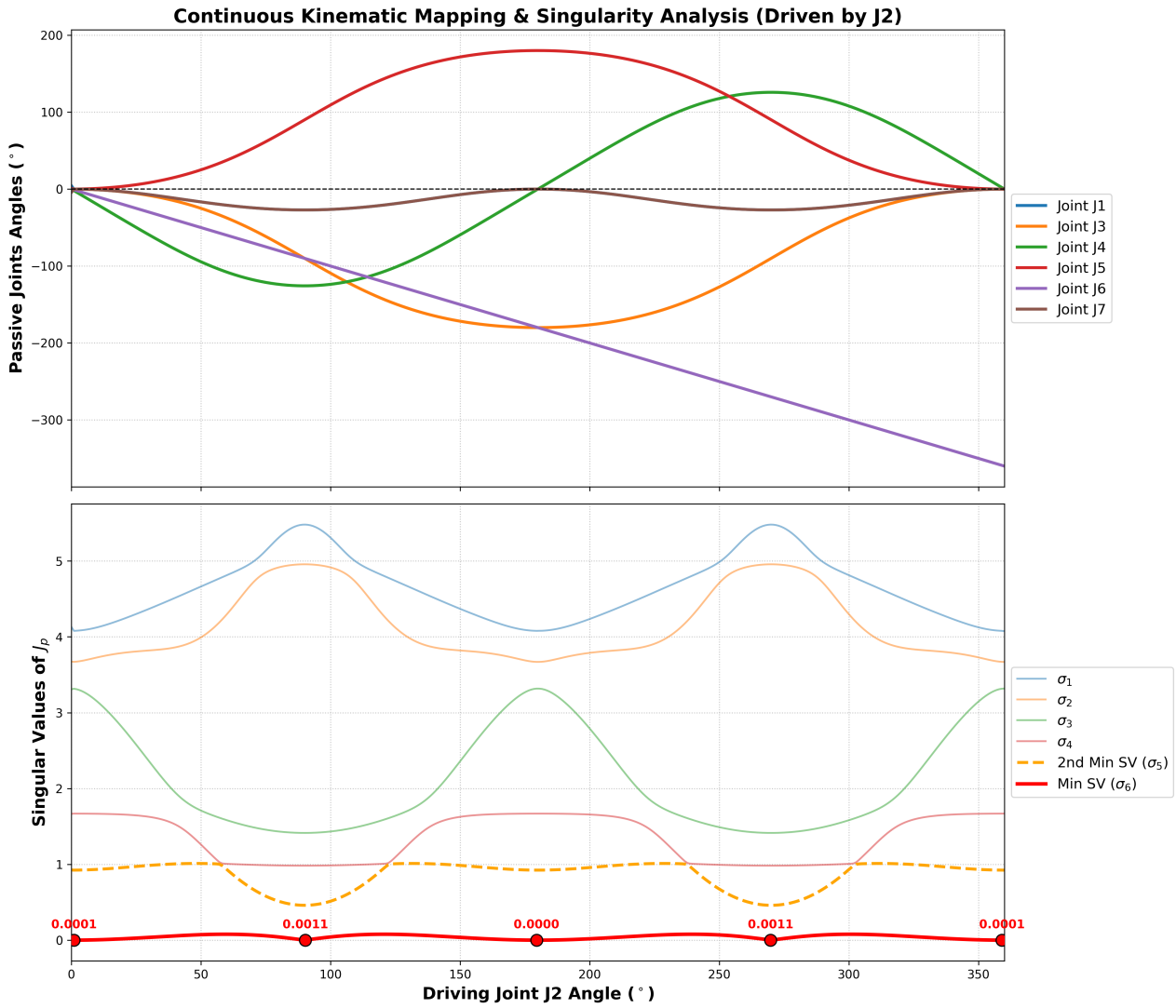


Figure 1: Continuous Kinematic Mapping & Singularity Analysis

2 Singularity Detection (Dead-center Points)

In numerical computation, a singularity occurs when the minimum singular value (σ_6) of the passive Jacobian matrix approaches zero. By finding the local minima of σ_6, the following critical singular configurations (where $\sigma_6 < 0.1$) were identified:

- Driving Joint $J_2 = 1.00°$, Minimum Singular Value $\sigma_6 = 0.000108$

- Driving Joint $J_2 = 90.25°$, Minimum Singular Value $\sigma_6 = 0.001138$
- Driving Joint $J_2 = 179.50°$, Minimum Singular Value $\sigma_6 = 0.000024$
- Driving Joint $J_2 = 269.75°$, Minimum Singular Value $\sigma_6 = 0.001138$
- Driving Joint $J_2 = 359.00°$, Minimum Singular Value $\sigma_6 = 0.000098$

3 Fourier Series Kinematic Mapping Formulas

The discrete kinematic numerical solutions for the passive joints (Q_i in degrees) driven by J_2 (x in radians) have been fitted to 3rd-order Fourier series:

- $J1(x) = -13.911 + 0.023\cos(x) - 0.000\sin(x) + 13.665\cos(2x) - 0.000\sin(2x) + 0.023\cos(3x) - 0.000\sin(3x)$
- $J3(x) = -89.997 + 94.670\cos(x) - 0.000\sin(x) + 0.006\cos(2x) - 0.000\sin(2x) - 4.989\cos(3x) + 0.000\sin(3x)$
- $J4(x) = -0.000 - 0.000\cos(x) - 124.037\sin(x) + 0.000\cos(2x) - 0.000\sin(2x) - 0.000\cos(3x) + 2.255\sin(3x)$
- $J5(x) = +90.001 - 94.661\cos(x) - 0.000\sin(x) + 0.002\cos(2x) + 0.000\sin(2x) + 4.998\cos(3x) - 0.000\sin(3x)$
- $J6(x) = -180.000 - 0.000\cos(x) + 114.589\sin(x) - 0.000\cos(2x) + 57.290\sin(2x) + 0.000\cos(3x) + 38.188\sin(3x)$
- $J7(x) = -13.933 - 0.020\cos(x) + 0.000\sin(x) + 13.620\cos(2x) + 0.000\sin(2x) - 0.020\cos(3x) + 0.000\sin(3x)$